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Operation and Troubleshooting of Closed Crankcase Ventilation (CCV) Systems for Automotive Natural Gas Engines

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Introduction

The intent of this document is to assist the field in determining actions to take with regard to contamination or accumulation of oil and moisture emulsion with a closed crankcase ventilation (CCV) system. The video shown below is an example of normal condensation from the EGR system.

Note : https://tsb.cumminsvirtualcollege.com/SB5579487_en.aspx
(https://tsb.cumminsvirtualcollege.com/SB5579487_en.aspx)

Emulsion Explained

Natural gas engines produce water as a natural byproduct of their combustion process. When the engine is cold the water vapor will condense on cold surfaces in the engine and CCV system. The condensed water vapor, when mixed with oil, will form an oil and water emulsion. Emulsion that is present in the CCV filter housing is not able to pass through the filter until it has separated back into water and oil. Separation of water and oil will occur naturally when the engine reaches normal operating temperatures. The presence of water droplets and moisture is normal in all CCV systems. The CCV system is designed to process and disperse liquid water adequately.

Some emulsion will also be found in 'cold' spots on the engine. These cold areas are usually thin walled covers, such as the valve cover and the gear cover, and oil filler caps that are found in these covers or on the ends of filler tubes. Such areas are much colder than the main mass of the engine and, in cold conditions, they will collect an accumulation of emulsion on their internal surfaces.

Emulsion in the CCV filter will increase pressure through the CCV filter and cause the CCV filter bypass to open, resulting in oil carry over to the intake system. This document provides guidelines to determine if excess emulsion is present and what actions are needed.

Inspection Guide Lines

The following guidelines will show differing levels of emulsion accumulation and provide advice to the maintenance technician on how to proceed. Different actions will be required depending on emulsion level. The actions are: none, monitor, and correction.

Definitions of Actions

Action: None

No corrective or monitoring actions are required. System maintenance should proceed according to the normal maintenance schedule.

Action: Monitor

The system is operating adequately but is to be monitored for any increase of emulsion deposit. The system is to be inspected at the next oil drain interval for any changes in condition.

Inspection of the CCV system is to be made at the end of an operational shift so that the engine will have been operating for some hours at normal operating temperatures. The system is **not** to be inspected cold or after short periods of operation where the engine does **not** reach operation temperature.

If the condition remains the same for three oil drain periods, or improves, no further monitor action is required.

Action: Correction

The CCV system is **not** adequately controlling the accumulation of emulsion. The following points are to be checked. Correction of any one or a combination of these points will improve CCV system operation and are to be monitored for improvement as described in the 'monitor' section above:

Condition 1: No Emulsion



Figure 1, Left: Valve Cover Filler Cap. Center: Valve Cover Filler Hole. Right: CCV Filter Bowl.

Table 1, No Emulsion Examples

Figure	Action	Notes
1	None	The emulsion is completely dispersed by the CCV system.

Condition 2: Light Emulsion



Figure 2, Left: Valve Cover Filler Cap. Center: Valve Cover Filler Hole. Right: CCV Filter Bowl.

Table 2, Light Emulsion Examples

Figure	Action	Notes
2	None	Emulsion deposits are normal and are being processed by the CCV system. The drain passages at the bottom of the filter bowl are not restricted. Emulsion in the oil filler cap is normal and there is no or very little visible emulsion inside the valve cover.

Condition 3: Medium Emulsion



Figure 3, Left: Valve Cover Filler Cap. Center: Valve Cover Filler Hole. Right: CCV Filter Bowl.

Table 3, Medium Emulsion Examples

Figure	Action	Notes
3	Monitor	Emulsion deposits are normal but are beginning to deposit on the interior wall of the CCV filter bowl. The drain passages of the CCV filter bowl are not restricted. Emulsion in the filler cap is opaque and there is visible emulsion inside the valve cover. This condition can either return to Condition # 2 or it may progress to Condition #4.

Condition 4: Heavy Emulsion



Figure 4, Left: Valve Cover Filler Cap. Center: Valve Cover Filler Hole. Right: CCV Filter Bowl.

Table 4, Heavy Emulsion

Figure	Action	Notes
4	Correction	Emulsion deposits are not under control. Oil filler cap emulsion is opaque and thick. Oil filler hole shows visible emulsion. This condition is unlikely to return to Condition #3 or Condition #2 unless operating conditions change or corrective actions are taken.

Inspection and Correction

Extended Engine Idle

During extended idle conditions, the engine is **not** under load and does **not** reach temperatures high enough to naturally disperse emulsion. If emulsion is found in the CCV system or other components after an extended idle, operate the engine under loaded conditions and allow it to reach normal operating temperatures before re-checking emulsion conditions.

In cold weather conditions, avoid operating the engine at an idle, no load condition for prolonged periods of time followed by a cold soak. If the engine is shut down and allowed to cold soak in temperatures below freezing, the emulsion and water in the CCV filter and housing can freeze restricting the operation of the CCV filter bypass. If the CCV filter and bypass are restricted the engine will have high crankcase pressure when operated. When operating under load, the engine becomes hot enough to naturally separate the water and oil emulsion allowing it to pass through the CCV filter.

If prolonged idling is unavoidable, the engine is to be idled at an rpm which is adequate to maintain coolant temperatures above 60°C [140°F]. At -18°C [0°F], the engine is to be idled above 1200 rpm. For prolonged idle in these cold weather conditions, a full on-off type fan and winter front **must** be used. The coolant thermostat, coolant thermostat seal, and the de-aerating system is **not** to allow coolant leakage through the radiator core with the coolant thermostat in the closed position.

Light load engine operation

Engines operated under light load may **not** reach high enough operating temperatures to naturally disperse emulsion. If emulsion is found after a light load operation, operate the engine under loaded conditions and allow the engine to reach normal operating temperatures before re-checking emulsion conditions.

Coolant Thermostat Operating Incorrectly

Check for any active fault codes related to coolant temperature and follow the appropriate troubleshooting tree(s). A malfunctioned coolant thermostat can cause excess emulsion formation by preventing the CCV system from reaching temperatures high enough to naturally disperse emulsion. In cold weather conditions, coolant works to heat and maintain engine temperatures. To achieve this, the coolant thermostat and coolant thermostat seals **must not** allow coolant to flow through the radiator when the coolant thermostat is closed. To check for coolant leakage past the thermostat seal and/or thermostat:

1. Remove the engine outlet hose at the coolant thermostat housing. Hold the engine outlet hose in the air above the radiator baffle level and run the engine against the governor. If water comes out of the water outlet connection, the coolant thermostat and/or coolant thermostat seal is leaking; often it is a malfunction of the coolant thermostat seal or debris keeping the coolant thermostat open.
2. If the coolant does **not** reach coolant thermostat opening temperature, changing the coolant thermostat will **not** correct the problem.

The second critical area for cold operating engines is coolant flow through the radiator core when the coolant thermostat is in the closed position. A properly functioning de-aeration system **must** be incorporated and there **must not** be reverse flow of coolant through either the fill line or the radiator. The de-aeration baffle **must not** leak and **must not** have vent holes. To check for this condition:

1. Drain the cooling system.
2. Remove the make up (fill line) at the water pump and plug the end.
3. Fill the area above the baffle with coolant. If water leaks through the drain cock of the radiator, remove the tank and repair. Coolant can be leaking through the core vent tube which

needs to be extended in the expansion area.

As mentioned above, if the coolant does **not** reach the coolant thermostat opening temperature, changing the coolant thermostat will **not** correct the issue. Further investigation is needed as to why the coolant is **not** reaching the coolant thermostat opening temperature.

When checking the coolant thermostat, inspect the coolant lines going to and from the CCV for any valves or kinks that would restrict coolant flow.

Coolant radiator fan engagement at low coolant temperatures (below coolant thermostat operating temperature)

If the radiator fan is operating continuously, or operating at low coolant temperatures, this can prevent the coolant from reaching the coolant thermostat opening temperature and cause emulsion to form in the system. Check for any active fault codes related to the radiator fan and follow the appropriate troubleshooting tree(s). Check the radiator fan on time and determine if the radiator fan is operating continuously and/or operating when the coolant temperature is below the coolant thermostat operating temperature. Additionally, check for any original equipment manufacturer (OEM) systems that may be requesting "fan on", such as the air conditioning system.

Cold ambient air flow over the engine or CCV system

The CCV system and all related hoses are to be routed to avoid being in the path of direct airflow. If any part of the CCV system is in the path of direct airflow, the temperature can be lowered enough to allow emulsion to form even if the engine is at normal operating temperatures. In cold weather conditions, this emulsion can freeze and lead to blockages. Reroute any CCV system hoses that are in the path of direct airflow or verify CCV system hoses are insulated if rerouting is **not** an option.

All CCV system hoses are to be routed so there are no bends or low points where water can collect. In cold weather conditions, water can freeze and cause a blockage and impair function. Reroute any CCV system hoses that have bends so moisture drains out of the CCV system as intended.

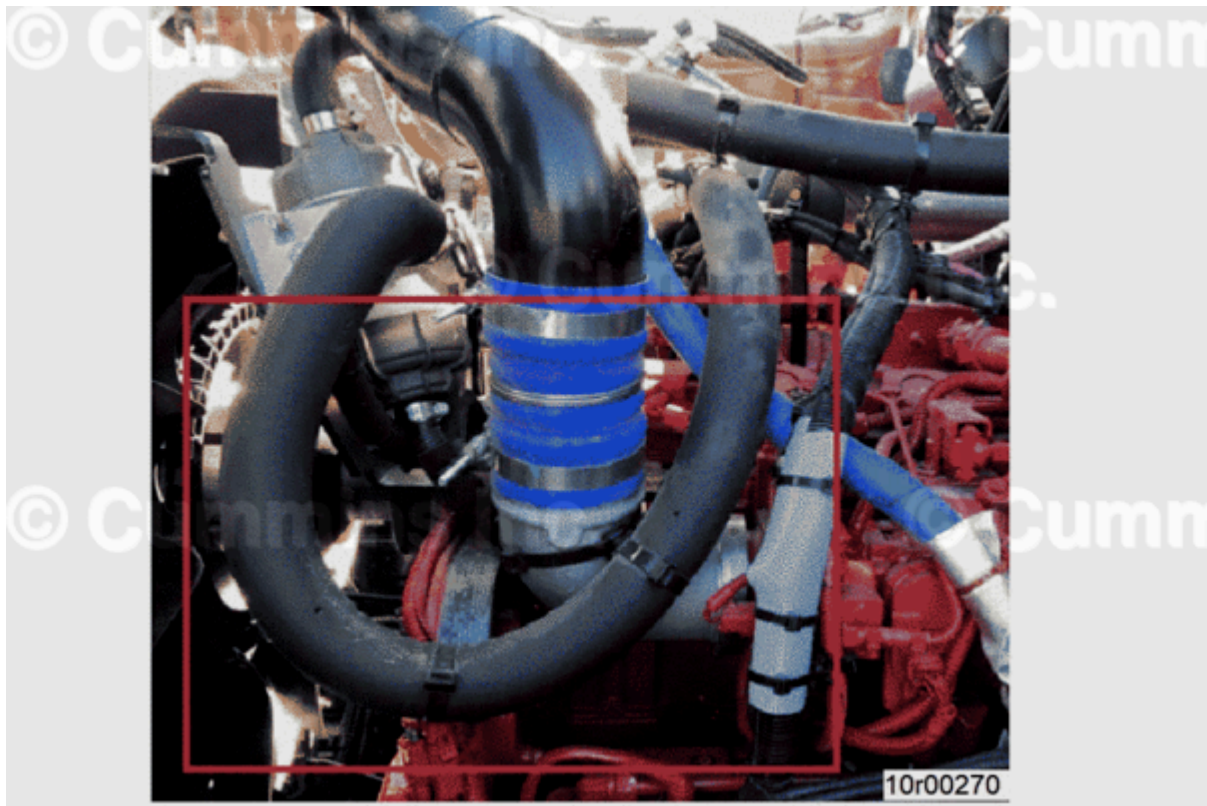


Figure 5, Improper CCV System Hose Routing Where Condensation Can Collect.

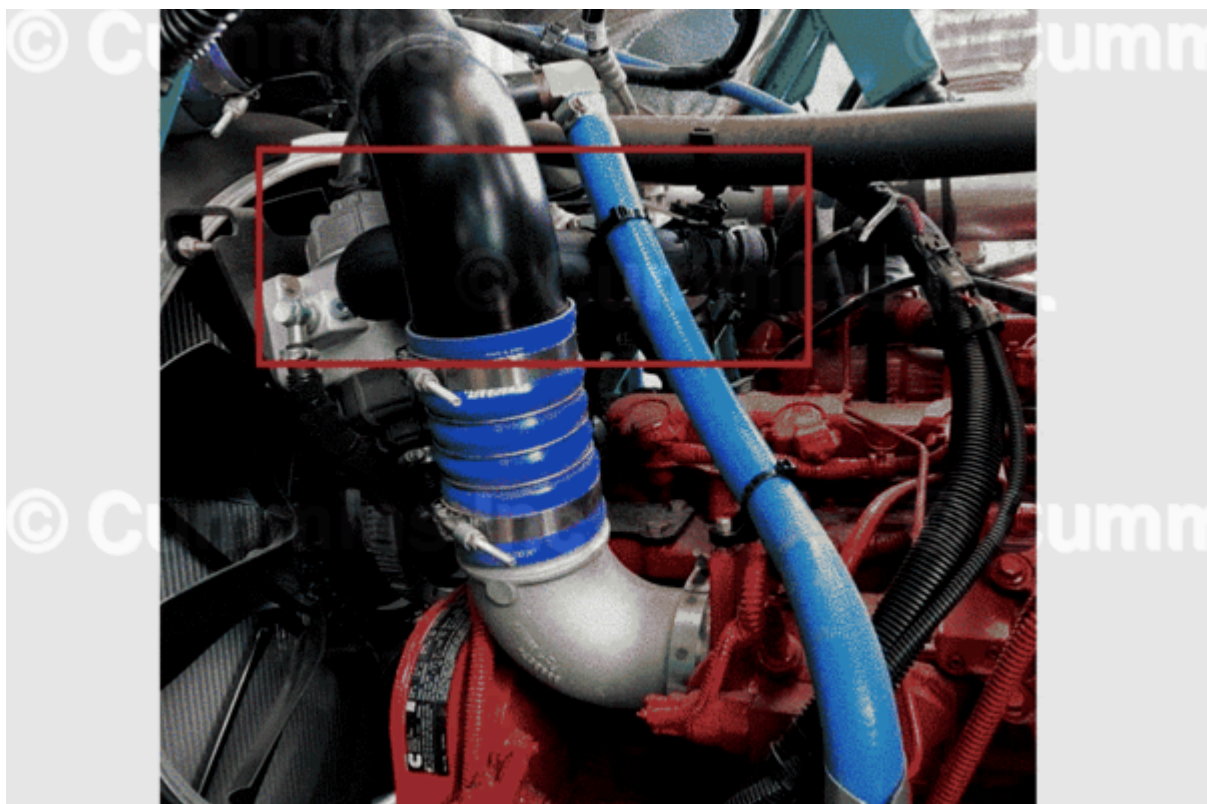


Figure 6, Proper CCV System Hose Routing So No Condensation Can Collect.

Improper insulation on CCV filter or hoses

Heated CCV systems are offered for ISL G Euro VI but are **not** required. If a non-heated CCV system is installed and the engine is operating in cold weather conditions, insulation jacket, Part Number 5399731, **must** be used.



Figure 7, Insulation Jacket, Part Number 5399731.

Oil Samples

Emulsion may also be visible in lubricating oil samples taken from an engine operating in cold climates with low loads or extended idle times, making lubricating oil samples appear cloudy and opaque. This is considered normal, because the water in the lubricating oil will begin to evaporate once the engine is driven at normal operating temperature. See Figures 8, 9, and 10, that show three acceptable lubricating oil samples from the lubricating oil rifle of a natural gas engine.

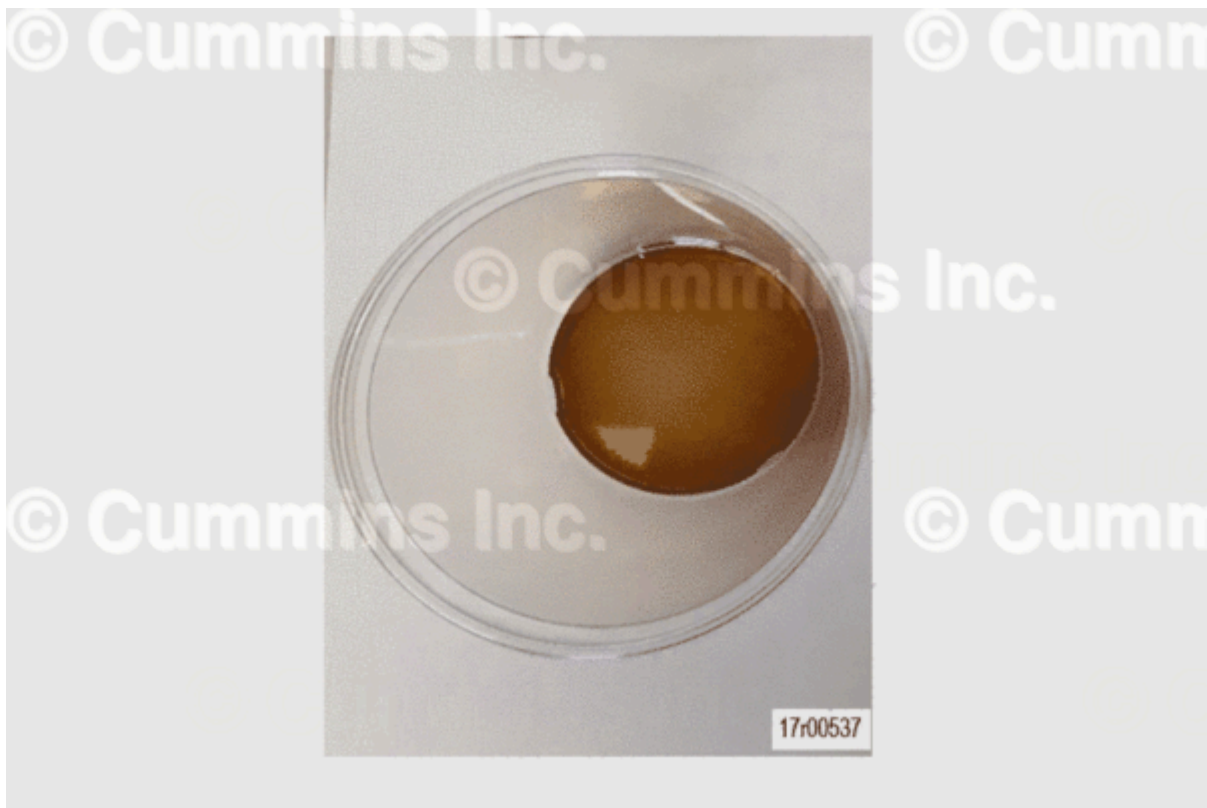


Figure 8, Lubricating Oil Sample Taken Shortly After A Lubricating Oil Change.



Figure 9, Lubricating Oil Sample Taken With Roughly Two Pints Of Water In The Crankcase Lubricating Oil.

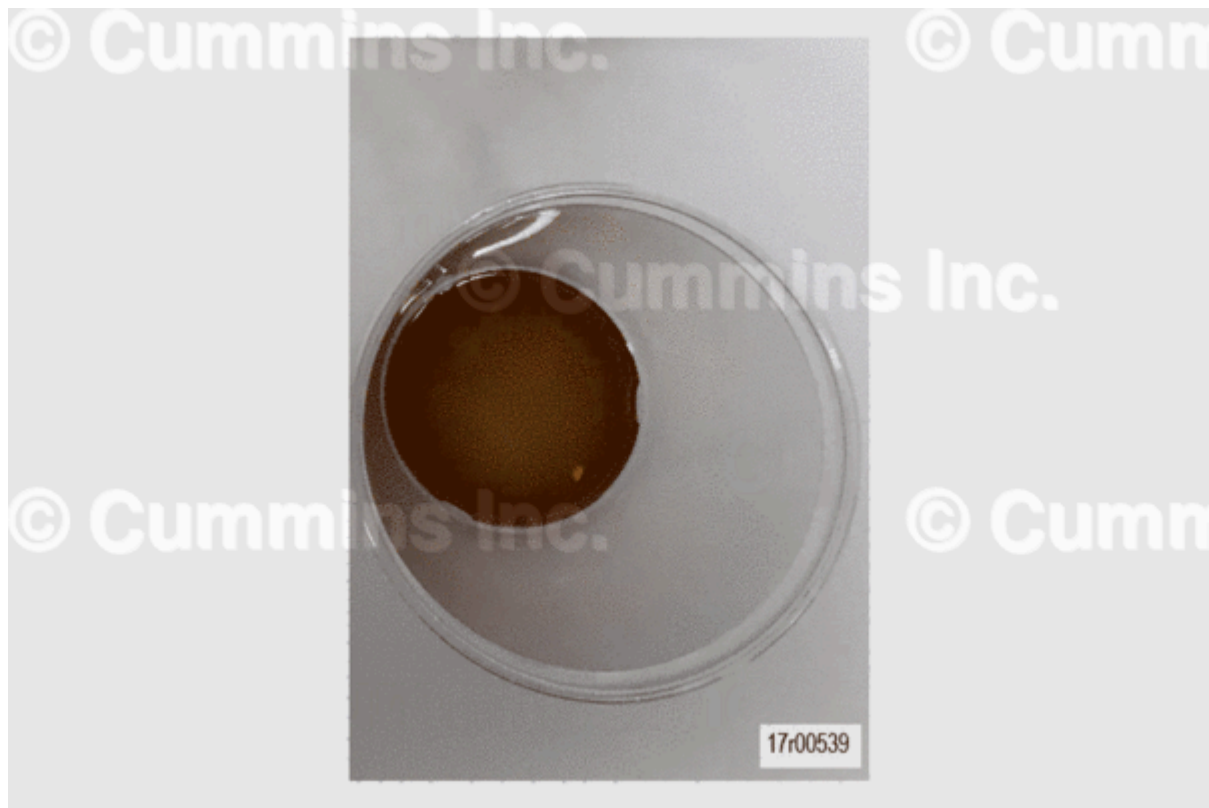


Figure 10, Lubricating Oil Sample Taken From Same Engine After Operating For Twenty Minutes.

Figure 8 above is a control sample taken shortly after a lubricating oil change, Figure 9 lubricating oil sample was taken with approximately two pints of water in the crankcase lubricating oil, and Figure 10 lubricating oil sample was taken after the unit was driven for twenty minutes. These lubricating oil samples were analyzed, and results showed that the initial and final lubricating oil samples had similar water concentrations that were below the 3000 parts per million specification listed in Section 378-013 of the Fluids for Cummins Products Service Manual, Bulletin 5411406. The water concentration in parts per million over the duration of this testing can be seen in Figure 11 below.

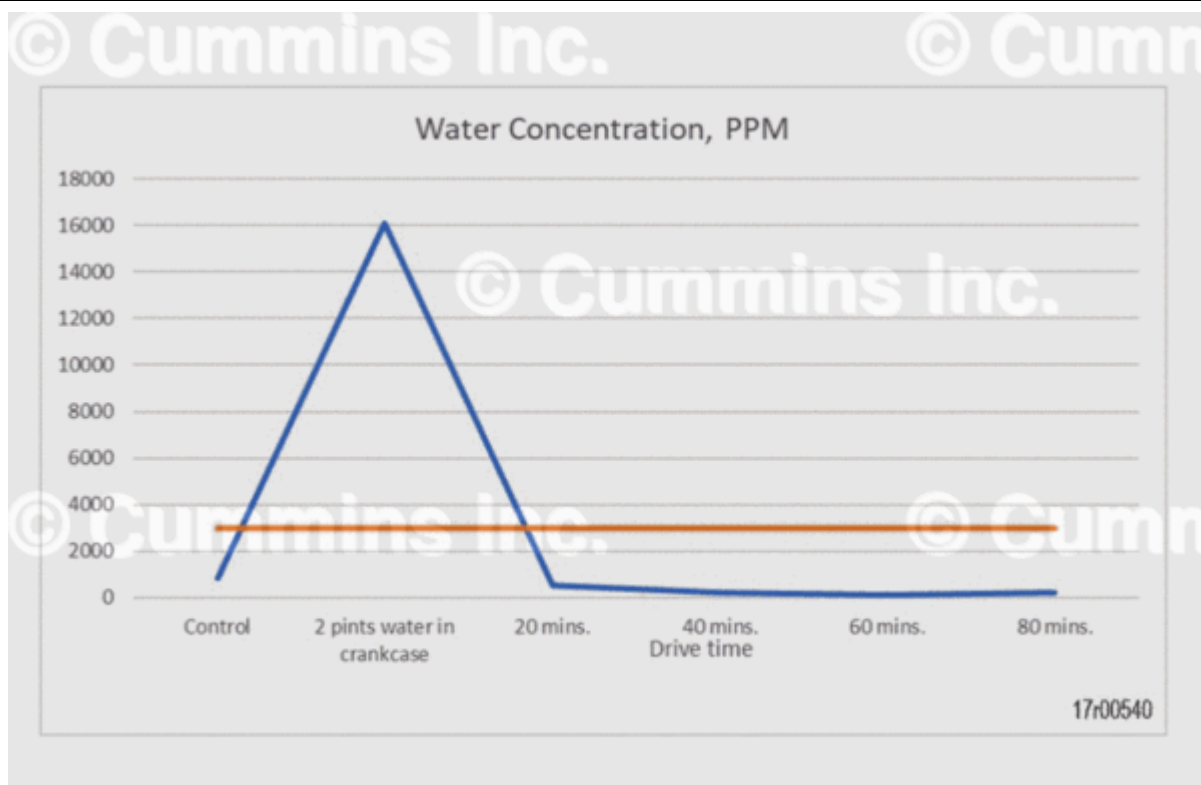


Figure 11, Results Of Water Dispersion Test Showing Maximum Allowable Water Concentration In Lubricating Oil Samples.

If a lubricating oil sample appears cloudy, testing must be performed to verify the sample does **not** contain engine coolant. If coolant is found in the lubricating oil sample, reference the appropriate symptom troubleshooting tree in the service manual for the engine. If coolant is **not** found in the lubricating oil sample, drive the vehicle with the engine at operating temperature and take another lubricating oil sample shortly after shutdown. Drive times required to evaporate water from the crankcase lubricating oil will vary based on operating conditions such as load and ambient temperature. If the engine will **not** reach operating temperature or if lubricating oil samples remain cloudy after operating the engine, reference the other sections of the Inspection and Correction portion of this Service Bulletin to verify proper engine operation in low ambient temperatures.

Other Considerations

Cylinder block and coolant heaters

In cold weather operation, cylinder block and coolant heaters bring the engine up to operating temperature faster and will aid in preventing and dispersing emulsion. For more information regarding cylinder block and coolant heaters, see Service Bulletin "Operation of Automotive Natural Gas Engines in Cold Climates, Bulletin 4332709".

Progressive Damage

High crankcase pressure caused by any of the conditions above can lead to the following symptoms:

Turbocharger oil leaks

High crankcase pressure can restrict turbocharger return oil flow, leading to engine oil leaks from the compressor housing and/or turbine housing. The turbocharger seal is not damaged by elevated crankcase pressure or oil leaks. Do not replace a turbocharger for oil leakage if CCV routing is incorrect or high crankcase pressure faults have logged. See corresponding Service Manual. Reference Procedure 010-033 in Section 10 for turbocharger diagnosis.

Gear cover damage

High crankcase pressure can cause oil leakage from the front gear cover seal, and/or front gear cover deformation. If gear cover appears to be damaged and front crank seal is leaking oil, inspect gear cover and seal for reuse. See corresponding Service Manual. Reference Procedure 001-031 in Section 1. Investigate and correct cause of high crankcase pressure.

Document History

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2018-9-6	Module Created
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